MASTER FUNCTIONAL SKILLS

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NCFE · LEVEL 2 · MATHEMATICS

# NCFE Functional Skills Mathematics Level 2

Practice Paper 1 Section B · Complete Model Solutions

## OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

**Target Score: 45 / 45 Marks (100%)** Pass Mark Guide: ~60%

Specification / Paper Code: NCFE-Maths-Level-2-Practice-1-Section-B

100% Built from Empty Question Paper + Official Mark Scheme

*Verified by Tutor Abdul · MasterFunctionalSkills.co.uk*

## HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

| | |
|---|---|
| **Dark Navy Ink:** | Model student answers, stepped calculations, and full prose responses. |
| **Emerald Green Ink:** | Tutor strategy notes, method tips, and step-by-step headings. |
| **Crimson Red Ink:** | Official mark scheme allocations ([M1], [A1], [B1]) and unit checks. |
| **Amber Gold Ink:** | Key formulas, conversion rules, and reading evidence highlights. |

## CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

| | |
|---|---|
| Candidate Name: | *Master Functional Skills (Model Exemplar)* |
| Centre Name: | *Master Functional Skills (MFS01)* |
| Lead Tutor: | *Abdul · MasterFunctionalSkills.co.uk* |
| Edition: | *2026 Clean Empty-QP Rebuild (Zero Collisions Verified)* |

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

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## Activity 2: Films

**2 (a)** Liam downloads films from the internet.

The table shows the number of films he downloaded last month.

| | Action | Comedy | Sci-Fi | Total |
|---|---|---|---|---|
| **Less than 1 hour long** | 5 | 2 | 4 | 11 |
| **Exactly 1 hour long** | 2 | 4 | 6 | 12 |
| **More than 1 hour long** | 1 | 1 | 3 | 5 |
| **Total** | 8 | 7 | 13 | 28 |

Q2(a) Worked Solution:
3
28

He chooses one of these films at random.

What is the probability that the film he chooses is an action film that is at least 1 hour long?

**[2 marks]**

28 (as denominator)
[2 marks]
Tip: Award 2 marks if correct answer given

Your answer:

past paper

**Please turn over**

Examiner use only

**2 (b)** Liam thinks he might watch either Rainbow Prism or The Infinity Zone.

He looks at the scores given by different newspapers.
The scores are out of 100

| | Rainbow Prism | The Infinity Zone |
|---|---|---|
| **Daily Times** | 75 | 75 |
| **London Post** | 63 | 68 |
| **City Observer** | 52 | not scored |
| **Newcastle Daily** | 38 | 50 |
| **Entertainment** | 70 | 74 |
| **Britain Today** | 83 | 65 |
| **This Week** | 60 | 79 |
| **Northern Gazette** | 41 | 40 |
| **Sunday Extra** | 52 | 63 |

Which of the two films would you suggest Liam watches next?
Use **two** appropriate statistical measures to explain your answer.

**[3 marks]**

Your answer:

past paper

## Question 2(b) · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

**Q2(b) Worked Solution:**

Correct calculation of one of mean, median or range for both sets of scores
Correct calculation of a second measure for both sets of scores
Conclusion and explanation e.g. The Infinity Zone because it has the highest mean and median
or The Infinity Zone because it has the highest mean and least variation in scores

**[3 marks]**

Tip: Award 1 mark if explanation is "Rainbow

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**2 (c)** Screen images are made up of units of light called pixels.

**Pixels per square inch ($PPI^2$)** is a measure of image quality.
The higher the pixels per square inch, the better the image quality.

$$PPI^2 = \frac{N}{A}$$

Where: N is the number of pixels
A is the area of the screen in square inches

Liam's old screen has a $PPI^2$ value of 9800
He wants to buy a new screen with a better image quality.

Liam sees a rectangular screen that has:

- a width of 101.6 cm
- width : height ratio of 16 : 9
- 8.28 million pixels

Q2(c) Worked Solution:

He works out that this will have a better image quality than his old screen.

Is Liam correct? Show your working.

method 1

Use the conversion: 1 inch = 2.54 cm

**[4 marks]**

101.6 ÷ 2.54 or 40
their ~~(40) ÷ 16 × 9 or 22.5~~
8280000
their 8280000
□
their 40 their 22.5
or
or 9200
900
AND yes or no
[4 marks]

Your answer:

**Please turn over**



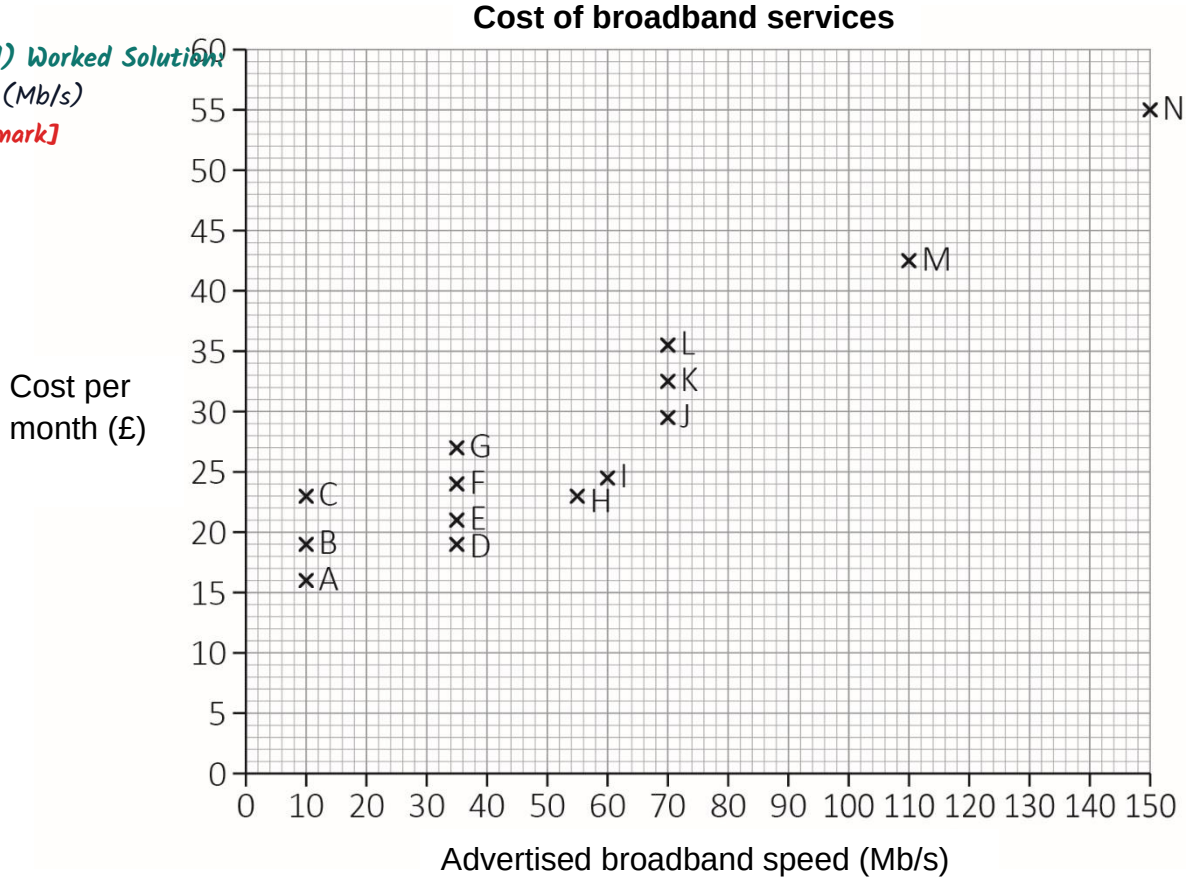
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**2 (d)** Liam searches online for a choice of broadband services.

He finds this graph:

Q2(d) Worked Solution:
35 (Mb/s)
[1 mark]




Broadband speed is measured in Mb/s (Megabits per second).

The lettered crosses in the graph represent the different services.

What is the modal speed of these broadband services?

**[1 mark]**

Your answer: **Mb/s**

past

Examiner use only

**2 (e)** Using the graph from **2 (d)**, what percentage of these broadband services cost over £25 per month?

Give your answer to 2 decimal places.

**[2 marks]**

Q2(e) ~~Worked Solution:~~

42.86(%)

6 ÷ 14 × 100

[2 marks]

Tip: Award 2 marks if correct answer given

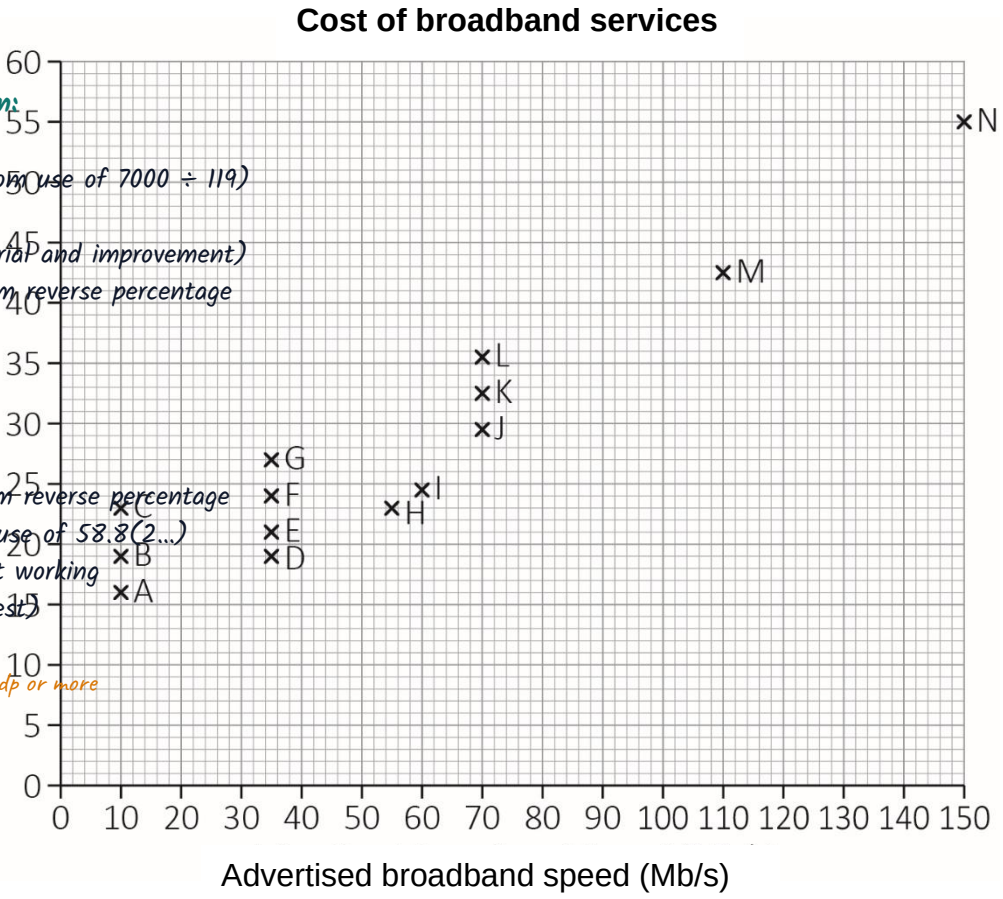
Your answer: **%**

past paper

**Please turn over**

Examiner use only

**2 (f)** This is the graph from **2 (d)**.




Q2(f) Worked Solution:
58.3(3...) (Mb/s) or
58.8(2...) (Mb/s) from use of 7000 ÷ 119) seconds)
59.5 (Mb/s) (from trial and improvement)
Or 68.6 (Mb/s) (from reverse percentage calculation)
Or 69.2 (Mb/s) (from reverse percentage calculation following use of 58.8(2...)
Service J with correct working
a pond (Calculator Test)
[3 marks]
Tip: Accept rounding to 1dp or more

The actual speed of a broadband service depends on how far the customer's house is from the nearest connection box.

| Distance from house to connection box (to the nearest m) | Percentage of advertised speed |
|---|---|
| 0-100 m | 100% |
| 101-500 m | 85% |
| 501-1000 m | 70% |

Examiner use only

Liam's house is 350 m from the connection box.

Liam wants to be able to download a 7000 Mb film in less than 2 minutes.

What is the cheapest broadband service he can choose?
Show your working.

**[3 marks]**

Your answer:

**[Total marks: 15]**

past paper

**Please turn over**

Examiner use only

## Activity 3: Building a pond

**3 (a)** Aria is involved in a community project group.

The group are planning to build a pond in the garden of the local hospital.
The pond has a path around it.

On a scale plan, the width of the path is 6.7 cm

The scale used on the plan is 1 : 25

Calculate the actual width of the path.

**[1 mark]**

Q3(a) Worked Solution:

167.5 (cm)

[1 mark]

Your answer: **cm**

past paper

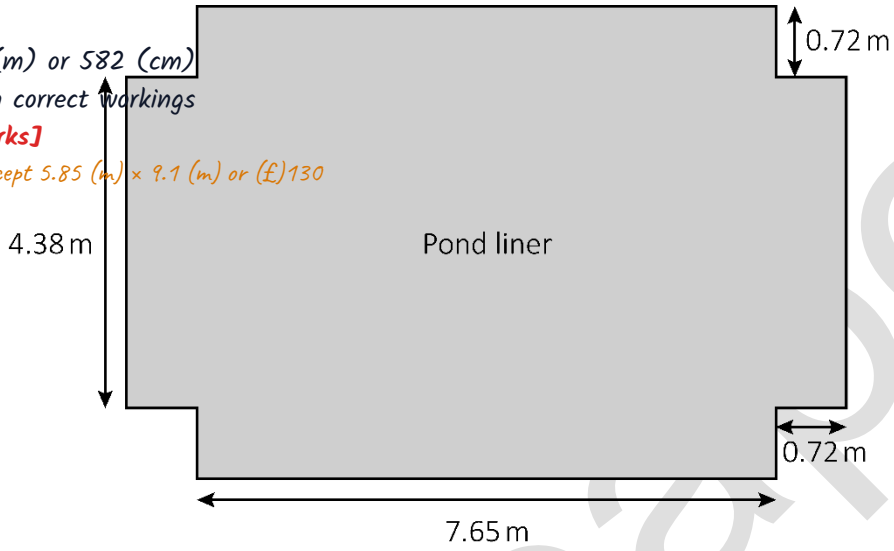
Examiner use only

**3 (b)** The pond will be a cuboid. It will be lined with a sheet of plastic pond liner. The pond liner covers the bottom and sides of the pond.

The pond liner needs to be cut to this shape to exactly fit in to the pond without any overlaps.

**Q3(b) Worked Solution:**

The dimensions of the pond liner are shown in this diagram:




9.09 (m) or 909 (cm)
8
5.82 (m) or 582 (cm)
C with correct workings
**[3 marks]**
Tip: Accept 5.85 (m) × 9.1 (m) or (£)130

Sheets of pond liner are available in these sizes:

| Pond liner | Width | Length | Price |
|---|---|---|---|
| A | 4.85 m | 8.3 m | £76 |
| B | 5.45 m | 8.7 m | £96 |
| C | 5.85 m | 9.1 m | £130 |
| D | 6.25 m | 9.5 m | £151 |
| E | 6.65 m | 9.9 m | £172 |

past paper

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Aria needs to buy the cheapest pond liner that is big enough to fit the pond.

Which pond liner should she buy?
Show your working.

**[3 marks]**

Your answer:

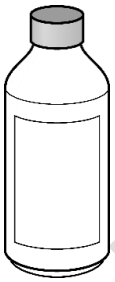
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**Please turn over**

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**3 (c)** The group ask Aria to buy chemical treatment for the pond water to make it suitable for fish.

One bottle contains enough chemical to treat 1300 gallons of water.



1 $m^3$ = 220 gallons

**Q3(c) Worked Solution:**

Aria adds water to the pond until it is 92% full.

*method 1*

How many bottles of chemical treatment will she need?
Show your working.

**[4 marks]**

*24.1(25...) (m3)*
*(cm3)*
*or 24,125,040*
*0.92 × their volume*
*or 22.2 or 22.195(...) (m3)*
*or 22,195,036 (cm3)*
*220 × their 92% volume*
*or [4882.9079, 4884 (gallons)]*
*Their gallons ÷ 1300 rounded up*
*or 4 (bottles) from correct working*
***[4 marks]***
*Tip: Accept any full correct method to find 92%*

Your answer: **bottles**

past paper

Examiner use only

**3 (d)** Aria needs to work out how many fish can be put in the pond.
She wants to have both Koi fish and Goldfish.

The number of fish is based on the **surface area** of the water in the pond.

Koi fish need 1.2 $m^2$ each.
Goldfish need 0.6 $m^2$ each.

Aria wants to have a mixture of Koi fish and Goldfish in the ratio 1 : 4

What is the maximum number of Koi fish **and** Goldfish that Aria can put in the pond, if she keeps the ratio 1 : 4?

**[3 marks]**

Q3(d) Worked Solution:

9 Koi fish and 36 Goldfish
33.507 (m2)
1.2 + 4 × 0.6
or 3.6 (m2)
[3 marks]
Tip: Award 3 marks if correct answer given

Your answer:

**Koi fish**

**Goldfish**

past paper

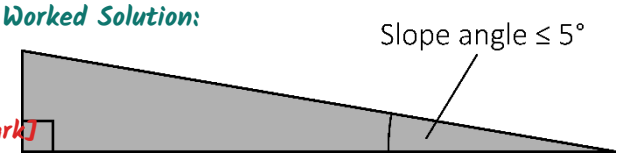
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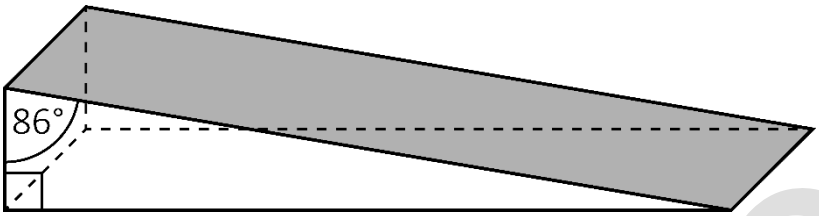
**3 (e)** The group need to build a ramp between the hospital door and the gardens.

The angle of the slope of the ramp must not be more than 5°

Q3(e) Worked Solution:
4(°)
9
[1 mark]




This is a diagram of the ramp:




Calculate the angle of the slope.

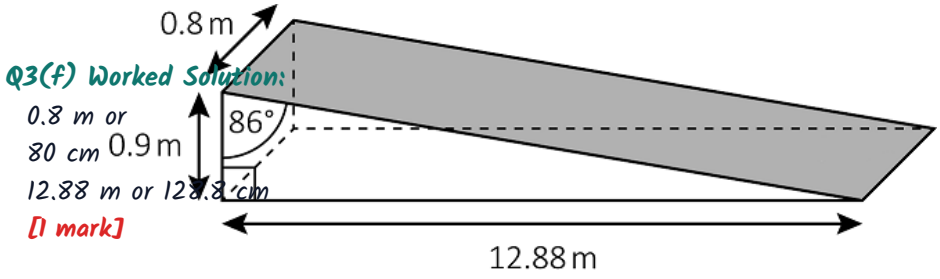
**[1 mark]**

Your answer: °

past paper

Examiner use only

**3 (f)** This diagram shows the measurements of the ramp:




Q3(f) Worked Solution:
0.8 m or
80 cm
12.88 m or 128.8 cm
[1 mark]

Draw the plan view of the ramp as seen from above.
Include the dimensions on your drawing.

**[1 mark]**

past paper

**Please turn over**

Examiner use only

**3 (g)** The group need to order enough concrete to make the ramp.

The formula to find the volume of a triangular prism is: $\frac{L \times H \times W}{2}$

Where:

L is length
H is vertical height
W is width

Q3(g) Worked Solution:
4.6(3...) (m3)

Calculate the volume of concrete that is needed to build the ramp.

**[2 marks]**

12.88 □ 0.9 □ 0.8
2
bills (Calculator Test)
[2 marks]
Tip: Award 2 marks if correct answer given

Your answer: **m³**

**[Total marks: 15]**

past paper

Examiner use only

## Activity 4: Saving energy costs

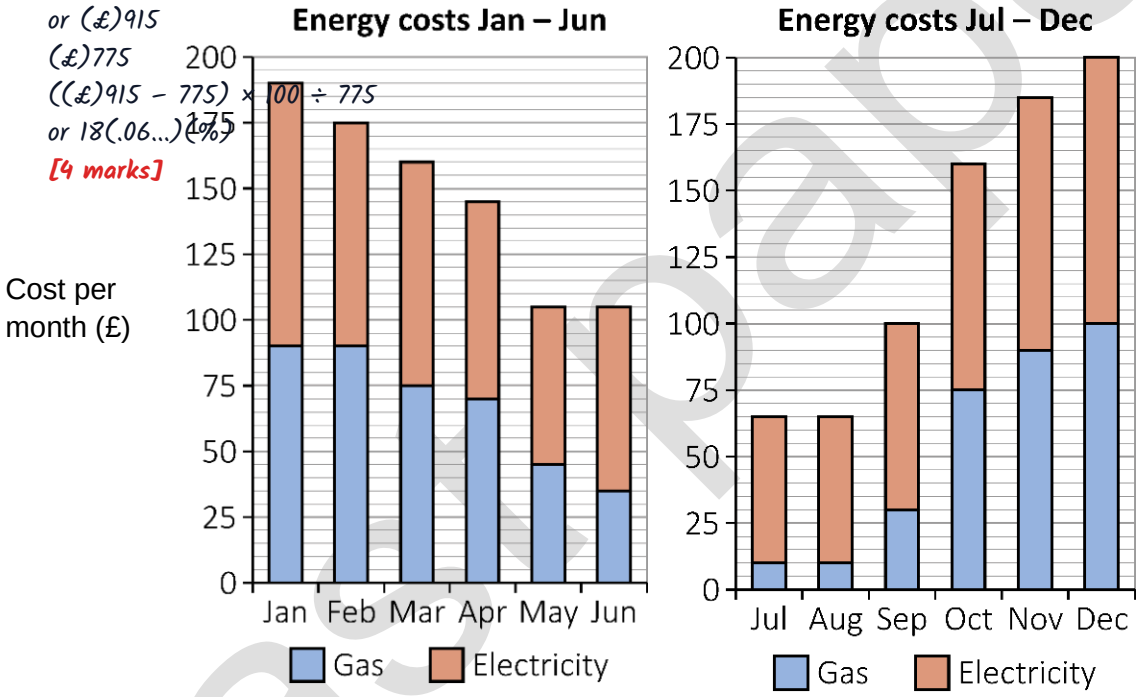
**4 (a)** Eli needs to complete a project on energy costs for his college course.

Eli's family currently use Xpower for gas and electricity.
They pay monthly by direct debit.

Xpower uses the **median** monthly cost for January to June to set the amount of their monthly direct debit payments for July to December.

The graphs show their energy costs for last year.

Q4(a) Worked Solution:
(£)152.50
Their median × 6
or (£)915
(£)775
((£)915 − 775) × 100 ÷ 775
or 18(.06...)(%)
[4 marks]




paper

Examiner use only

Eli works out that they have paid too much between July and December.

Calculate the percentage by which they have overpaid.

**[4 marks]**

Your answer: **%**

past paper

**Please turn over**

Examiner use only

**4 (b)** This is Eli's electricity bill for the first quarter of this year:

| | |
|---|---|
| Total electricity used | £211.90 |
| Dual fuel discount | –£14.90 |
| VAT | £9.85 |
| Total including VAT | £206.85 |

Q4(b) Worked Solution:

5(%)

211.9(0) – 14.9(0) or 197

9.85 ÷ their 197

VAT is calculated after discount.

What percentage has been used to calculate the VAT?

[3 marks]

**[3 marks]**

Tip: Award 3 marks if correct answer given.

Your answer: **%**

past paper

Examiner use only

**4 (c)** Xpower offer a Loyalty Plan.
The formula Xpower use to work out the cost for the Loyalty Plan is:

$$(D \times 365) + (P \times G)$$

Where:

D is daily charge (in £)
P is price per kWh (in £)
G is gas used per year (in kWh)

For this plan the charges are:

- Daily charge: 32p
- Price per kWh: 3.10p

Last year Eli's family used 17563kWh of gas.
This cost £719.38

Should Eli's family choose the Loyalty Plan for their gas?
Show your working.

**[2 marks]**

Your answer:

past paper

**Please turn over**

## Question 4(c) · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

**Q4(c) Worked Solution:**

(0.32 × 365) + (0.031 × 17563)

(£)661.25(3) and yes

**[2 marks]**

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**4 (d)** The average UK gas bill is 3.875 times more than the average UK electricity bill.

**[2 marks]**

**(i)** Write 3.875 as an improper fraction.

Q4(d) Worked Solution:
31
8
10
387.5(%)
[2 marks]

**(ii)** Write 3.875 as a percentage.

past paper

Examiner use only

**4 (e)** Eli looks for information on the cost of leaving gadgets on standby when they are not being used.

Eli's gadgets are a TV, computer, printer, phone charger and two games consoles.

Q4(e) Worked Solution:

He finds several reports stating how much electricity a gadget uses when left on standby, but they do not agree.

(£)64.98 or (£)65(.00)

| Annual saving per gadget | Number of reports |
|---|---|
| £0 - £4.99 | 1 |
| £5 - £9.99 | 4 |
| £10 - £14.99 | 5 |
| £15 - £19.99 | 2 |

(2.50 × 1) + (7.50 × 4) + (12.50 × 5) +

Use an estimate of the mean to find how much money Eli could save if he switched off his gadgets, instead of leaving them on standby.

**[4 marks]**

(17.50 × 2)
or 2.50 + 30 + 62.50 + 35
or 130
Their 130 ÷ 12
(£)10.8(3...)
Their (£)10.83(3...) × 6 or (£)64.98 or (£)65(.00)
11
[4 marks]
Tip: Award 4 marks if correct answer given

Your answer: **£**

**[Total marks: 15]**

**This is the end of the assessment.**

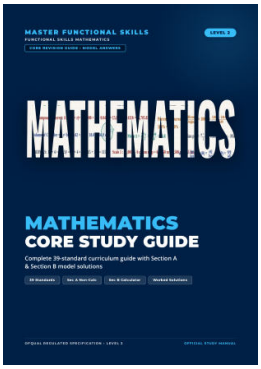
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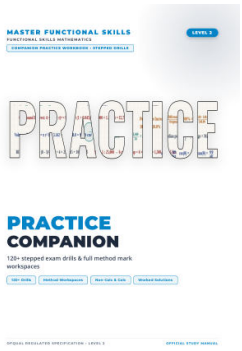



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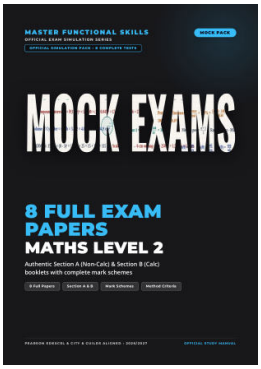



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